

WP5.3 — Deliverables with Sub-Tasks

D1 — Data Ingestion & Dataset Preparation

Objective

Prepare modeling-ready datasets from WP5.1 or sample setups.

Sub-Tasks

- Define required process signals and KPIs
- Identify input/output variables for SI
- Define data schema and naming conventions
- Develop data ingestion pipeline/interface
- Synchronize timestamps across signals
- Handle missing/noisy data
- Data filtering and preprocessing
- Feature extraction and normalization
- Dataset labeling (defects/events)
- Generate train/validation/test datasets
- Dataset quality assessment

Output

- Structured modeling-ready dataset
-

D2 — System Identification Framework

Objective

Develop behavioral/process model from acquired data.

Sub-Tasks

- Define modeling strategy
 - physics-based
 - ML-based
 - hybrid modeling
- Select SI tools/frameworks
 - MATLAB SI toolbox
 - PyTorch/TensorFlow
- Define model inputs/outputs
- Implement baseline models
- Train and calibrate models
- Hyperparameter tuning
- Compare modeling approaches
- Build reusable SI workflow
- Model version management

Output

- Identified behavioral model
-

D3 — Offline Model Validation Framework (Validation Loop 1)

Objective

Validate identified model before deployment/use.

Sub-Tasks

- Train/test split definition
- Define validation KPIs
- Compute prediction accuracy metrics
- Residual/error analysis
- Sensitivity testing

- Stability checks
- Model confidence estimation
- Validation reporting
- Define model acceptance criteria

Output

- Validated behavioral model
 - Validation metrics/report
-

D4 — Behavioral Digital Twin Framework

Objective

Use identified model for virtual process simulation.

Sub-Tasks

- Define DT architecture
- Integrate identified model into DT
- Build process simulation workflow
- Define virtual process inputs
- Develop prediction pipeline
- Implement scenario execution framework
- Simulate process behavior
- Generate predicted outputs
- KPI prediction framework
- Define DT interfaces/APIs

Output

- Behavioral digital twin environment
 - Predicted process outputs
-

D5 — Process Intelligence, Simulation & Optimization Framework

Objective

Provide framework for simulation, process analysis, and optimization studies.

Sub-Tasks

- Develop what-if simulation capability
- Parameter variation studies
- Sensitivity analysis workflow
- KPI correlation framework
- Root-cause analysis workflow
- Bottleneck analysis framework
- Define optimization objectives
- Define optimization constraints
- Implement optimization workflow
- Parameter recommendation logic
- Multi-objective optimization studies
- Process insight generation
- Reporting/dashboard interfaces

Output

- Simulation & optimization framework
 - KPI insights
 - Optimization recommendations
-

D6 — Virtual Commissioning Integration

Objective

Validate process and recommendations virtually.

Sub-Tasks

- VC environment integration
- Process feasibility checks
- Constraint validation
- Virtual process execution
- Collision/reachability verification
- Validate recommended parameter sets
- Generate VC validation reports
- Integration with DT outputs

Output

- Virtual validation results
-

D7 — WP5.1 Visualization Interface

Objective

Send predictive intelligence outputs to WP5.1.

Sub-Tasks

- Define interface/API with WP5.1
- Define prediction output schema
- Define KPI visualization data structure
- Export predicted outputs
- Export model health indicators
- Export optimization insights
- Alarm/alert generation support
- Dashboard integration support
- Traceability metadata support

Output

- Visualization-ready predictive outputs
-

D8 — Online Validation & Continuous Learning Framework

Objective

Continuously improve model using runtime data.

Sub-Tasks

- Predicted vs actual comparison workflow
- Runtime error computation
- Drift detection framework
- Model performance monitoring
- Retraining trigger logic
- Incremental learning workflow
- Dataset update mechanism
- Re-validation workflow
- Continuous improvement pipeline

Output

- Continuous learning workflow
 - Model improvement indicators
-

D9 — MES / Traceability Interface Support

Objective

Provide integration interfaces for MES and traceability.

Sub-Tasks

- Define MES integration interfaces
- Define optimization result export format
- Store process metadata

- Store model metadata
- Traceability record management
- Process history linkage
- Reporting interface support

Output

- MES/traceability integration support
-

Very Useful Management Summary

Deliverable	Main Focus
D1	Data preparation
D2	Model development
D3	Model validation
D4	Digital twin simulation
D5	Analysis & optimization framework
D6	Virtual validation
D7	Monitoring integration
D8	Continuous learning
D9	MES & traceability support





These are the approximate dates and can. You give the subtasks date aligning with these

D1 — Modeling Data Preparation (16-Jun-2026 → 21-Aug-2026)

D1.1 Sample mTorres Data Collection & KPI Identification

Subtasks

- Identify required process signals and KPIs
- Define data schema and tags
- Collect historical/sample machine data
- Validate signal availability
- Create initial dataset structure

Timeline

16-Jun-2026 → 10-Jul-2026

D1.2 Data Quality & Preprocessing

Subtasks

- Timestamp synchronization
- Missing data handling
- Noise filtering
- Outlier detection
- Data normalization
- Feature extraction

Timeline

11-Jul-2026 → 05-Aug-2026

D1.3 Modeling Dataset Finalization

Subtasks

- Generate train/validation/test datasets
- Dataset labeling
- Final data validation

- Curated dataset release

Timeline

06-Aug-2026 → 21-Aug-2026

D2 — Modeling Data Processing from WP5.1 (1-Oct-2026 → 22-Jan-2027)

D2.1 WP5.1 Interface Definition

Subtasks

- Define data exchange interface
- Define ingestion APIs/connectors
- Define data formats/schema

Timeline

01-Oct-2026 → 25-Oct-2026

D2.2 Data Ingestion Pipeline Development

Subtasks

- Build ingestion workflow
- Implement synchronization logic
- Implement preprocessing pipeline
- Logging/error handling

Timeline

26-Oct-2026 → 10-Dec-2026

D2.3 Dataset Structuring & Validation

Subtasks

- Dataset quality checks
- Curated dataset generation
- Validation against WP5.1 signals
- Documentation

Timeline

11-Dec-2026 → 22-Jan-2027

D3 — Defect / Design Model Development (1-Apr-2027 → 18-Jun-2027)

D3.1 KPI / Process Correlation Study

Subtasks

- KPI interaction analysis
- Process variable mapping
- Initial process understanding
- Correlation studies

Timeline

01-Apr-2027 → 22-Apr-2027

D3.2 Defect / Process Mapping

Subtasks

- Defect-to-process mapping
- Label refinement
- Root-cause interaction study
- Sensitivity identification

Timeline

23-Apr-2027 → 18-May-2027

D3.3 Feature Engineering & Dataset Refinement

Subtasks

- Feature selection
- Derived feature generation
- Dataset balancing
- Curated training dataset creation

Timeline

19-May-2027 → 18-Jun-2027

D4 — System Identification with Simulated Data (3-Aug-2026 → 4-Dec-2026)

D4.1 SI Problem Definition

Subtasks

- Define model inputs/outputs
- Define target process behavior
- Select SI methodology

Timeline

03-Aug-2026 → 28-Aug-2026

D4.2 SI Framework Setup

Subtasks

- MATLAB/Python SI environment setup
- Define reusable workflows
- Data connectors setup

Timeline

29-Aug-2026 → 25-Sep-2026

D4.3 Initial Behavioral Model Development

Subtasks

- Baseline model creation
- Parameter estimation
- Training/calibration
- Initial prediction validation

Timeline

26-Sep-2026 → 30-Oct-2026

D4.4 Offline Validation Loop 1

Subtasks

- Train/test validation
- Residual analysis
- Error metrics computation
- Model tuning

Timeline

31-Oct-2026 → 20-Nov-2026

D4.5 Final Behavioral Model Release

Subtasks

- Documentation
- Final validation
- Export reusable model
- Model readiness review

Timeline

21-Nov-2026 → 04-Dec-2026

D5 — System Identification Improvement with 5.1 Data (19-Jun-2027 → 3-Sep-2027)

D5.1 Real Data Integration

Subtasks

- Integrate WP5.1 datasets
- Align real signals with SI workflow
- Dataset synchronization

Timeline

19-Jun-2027 → 10-Jul-2027

D5.2 Model Retraining

Subtasks

- Retrain behavioral model
- Compare simulated vs real behavior
- Improve robustness

Timeline

11-Jul-2027 → 05-Aug-2027

D5.3 Validation Loop 2 (Online Validation)

Subtasks

- Predicted vs actual comparison
- Runtime error monitoring
- Drift analysis
- Model improvement workflow

Timeline

06-Aug-2027 → 20-Aug-2027

D5.4 Final Improved Model

Subtasks

- Final validation
- Performance reporting
- Release updated behavioral model

Timeline

21-Aug-2027 → 03-Sep-2027

D6 — Analysis Framework & Digital Twin (1-Oct-2026 → 19-Mar-2027)

D6.1 Behavioral Digital Twin Integration

Subtasks

- Integrate identified model into DT
- Define DT execution pipeline
- Define simulation interfaces

Timeline

01-Oct-2026 → 10-Nov-2026

D6.2 Scenario Simulation Framework

Subtasks

- What-if simulation capability
- Parameter variation studies
- Scenario execution logic

Timeline

11-Nov-2026 → 20-Dec-2026

D6.3 KPI Correlation & Insight Framework

Subtasks

- KPI correlation workflow
- Sensitivity analysis
- Root-cause analysis framework

Timeline

21-Dec-2026 → 31-Jan-2027

D6.4 Optimization Framework

Subtasks

- Define optimization objectives
- Constraint definition
- Recommendation logic
- Multi-objective optimization studies

Timeline

01-Feb-2027 → 28-Feb-2027

D6.5 WP5.1 Visualization Integration

Subtasks

- Predicted output interface
- Model health visualization
- KPI insight export
- Dashboard integration support

Timeline

01-Mar-2027 → 19-Mar-2027

Actually from 5.3 we don't have to do any of these Timestamp synchronization Missing data handling Noise filtering Outlier detection Data normalization Feature extraction This will be done by 5.1 Once we ingest data , we need to prepare the data for system identification purposes

Then your scope becomes much cleaner and more accurate architecturally.

Those activities:

- timestamp synchronization
- filtering
- noise handling
- normalization
- preprocessing

belong to:

WP5.1 — Data Processing / Monitoring Layer

NOT WP5.3.

WP5.3 should assume:

“cleaned and curated process data is available”

Then WP5.3 focuses on:

Modeling-oriented preparation

for:

- SI
- DT
- optimization

Corrected WP5.3 Scope

Instead of:

Data preprocessing

Use:

Modeling dataset preparation

Corrected D1

D1 — Modeling Data Preparation

Objective

Prepare WP5.1 curated data for System Identification workflows.

Correct Subtasks

D1.1 Define SI Modeling Variables

- Define model inputs/outputs
 - Define KPIs for SI
 - Define target process behavior
 - Define process states/labels
-

D1.2 Modeling Dataset Structuring

- Select relevant process signals
 - Windowing/sequence preparation
 - Train/validation/test split
 - Dataset formatting for MATLAB/Python SI frameworks
 - Scenario grouping
-

D1.3 Feature & Label Preparation for SI

- Derived feature preparation for modeling
 - Defect/event labeling for learning
 - Behavioral state classification
 - Dataset balancing
-

D1.4 Dataset Validation for SI Readiness

- Validate completeness for modeling
 - Validate SI compatibility
 - Validate model coverage
 - Curated modeling dataset release
-

MUCH BETTER ARCHITECTURE

WP5.1	WP5.3
Data acquisition	System identification
Signal synchronization	SI dataset preparation
Filtering	Behavioral modeling
Noise removal	Digital twin
Preprocessing	Simulation
KPI calculation	Optimization

This separation is architecturally correct.

Important wording change

Instead of:

Data preprocessing

Use:

Model-oriented dataset preparation

or

Dataset preparation for behavioral modeling

Updated realistic subtasks

What WP5.3 should actually do

Inputs from WP5.1

Clean synchronized process dataset

WP5.3 tasks

Select → structure → label → prepare
for SI and DT

NOT:

Acquire → filter → clean

Best concise wording for roadmap

Deliverable

Modeling dataset preparation

Output

SI-ready behavioral modeling dataset

This is probably the most accurate phrasing for your scope.

TASKS

Your structure is actually quite strong already. The overall sequence is technically logical and much better than a typical high-level R&T roadmap.

You already have:

- architecture phase
- sample-data phase
- SI phase
- DT phase
- optimization phase
- real-data refinement phase
- VC phase
- referencing phase
- MES phase

which is a very good decomposition.

But yes — there are:

1. some repetitions
2. some misplaced tasks
3. some missing engineering tasks
4. some naming inconsistencies
5. some deliverable overlaps

Below is a detailed review.

1. Biggest Issue — D Number Reuse / Conflicts

You reused:

- D4
- D5

- D6
- D7
- D8

for multiple independent domains.

This will become very confusing later during:

- reviews
- audits
- milestone tracking
- presentations
- Jira linkage

Recommendation

Keep separate numbering domains.

Example:

Domain	Suggested Prefix
HDT/SI	HDT-Dx
Optimization	OPT-Dx
VC	VC-Dx
Referencing	REF-Dx
MES	MES-Dx

OR simpler:

Current	Better
D4 Ver1 SI	D4
D5 Ver2 SI	D5
D6 Ver1 Analysis	D6
D7 Ver2 Analysis	D7
D8 Visualization	D8
D6 Optimization	D9
D7 Real-time prediction	D10
D8 VC	D11

Otherwise people will get lost.

This is the MOST important cleanup.

2. D2 is GOOD overall

D2 structure is actually very solid.

You correctly separated:

- signal understanding
- SI variable planning
- dataset structuring
- feature/label prep
- validation

Very good.

3. D2.5 — Slightly Too Detailed

This section is technically excellent, but roadmap/task sheets should not contain too much theory.

Example:

Power-to-Force ratios to capture actual physical load...

Too deep for milestone planning sheet.

Move those explanations into:

- technical note
- architecture doc
- methodology doc

Keep roadmap concise.

Recommendation

Shorten to:

D2.5 Feature & Label Preparation for SI

- Derived feature generation
- Defect/event labeling
- Behavioral state classification

- Dataset balancing
- Time-delay feature alignment

Much cleaner.

4. D3 — Structure Needs Cleanup

Currently D3 mixes:

- ingestion architecture
- defect mapping
- final dataset preparation

These are different maturity levels.

Recommended D3 Structure

D3 — Data Ingestion from WP5.1 — Design & Development

D3.1 Interface Contract Definition

- schema
 - metadata
 - naming
 - APIs
 - signal definitions
-

D3.2 Ingestion Pipeline Development

- ingestion framework
- parser

- storage
 - versioning
 - simulator adapter
-

D3.3 WP5.1 Integration Validation

- dry-run integration
 - schema validation
 - stability validation
-

Then move:

- defect mapping
- modeling-ready dataset preparation

to D4.

This is cleaner architecturally.

5. D4 (Actual laying data) — Missing Tasks

D4 currently almost empty.

You need the tasks we discussed.

Recommended D4 Structure

D4 — Actual Laying Data Integration & Modeling Preparation

D4.1 Actual Data Reception

- receive actual laying data
 - validate completeness
 - dataset registration
-

D4.2 SI Variable Refinement

- validate SI inputs/outputs
 - refine KPIs
 - refine process states
 - refine prediction targets
-

D4.3 Modeling Dataset Preparation

- train/validation/test preparation
 - sequence generation
 - scenario grouping
 - DT datasets
-

D4.4 Defect-to-Process Mapping

- align defects to process windows
 - initial root-cause mapping
-

D4.5 Dataset Validation

- SI compatibility
- DT compatibility
- prediction coverage validation

This is one of the biggest missing sections.

6. D4 Ver1 SI — Missing Important Engineering Tasks

You have:

- develop SI
- system mapping
- algorithms

Good.

But missing:

- baseline model comparison
 - model calibration
 - residual analysis
 - offline validation loop
-

Add

D4.x Offline Validation Loop 1

- train/test validation
- residual analysis
- error metrics
- model tuning

Very important technically.

7. D5 Ver2 SI Improvement — GOOD

Actually good structure.

No major issues.

Maybe add:

- drift analysis
 - online validation
-

Add

D5.x Runtime Model Monitoring

- prediction vs actual
- drift monitoring
- retraining trigger logic

Because this becomes your:

Validation Loop 2

8. D6/D7 Analysis + DT — Some Repetition

You currently split:

- Ver1 with sample data
- Ver2 with actual data

This is okay.

But:

- KPI correlation
- sensitivity
- RCA

are repeated.

Better Structure

D6 Ver1

Develop framework with sample data

D7 Ver2

Validate/refine framework using actual data

NOT redevelop.

Rename D7 Tasks

Instead of:

Framework for KPI correlation analysis

Use:

Validation and refinement of KPI correlation framework using actual laying data

This avoids repetition.

9. Optimization Section — Missing Constraints Definition

Optimization without constraints is dangerous.

Add:

Optimization Constraint Framework

- process limits
- machine limits
- safety constraints
- feasible operating windows

Very important industrially.

10. Real-time Prediction Section — VERY GOOD

Actually strong.

Maybe rename:

Predictive indicators

to:

Early warning indicator framework

More professional.

11. VC Section — Missing Digital Twin Connection

Currently VC appears isolated.

Need explicit linkage:

Add

Integration of behavioral DT outputs with VC validation

Otherwise reviewers may ask:

How VC connects to HDT?

12. Referencing Section — VERY GOOD

Actually one of the cleanest sections.

Only missing:

Add

Reference frame transformation validation

between:

- localization
 - NC correction
-

13. MES Section — Too High Level

Need:

- traceability
 - storage
 - metadata
 - optimization history
-

Add

MES Data Mapping

- optimization result storage
 - DT metadata storage
 - traceability mapping
-

14. BIGGEST Missing Deliverable

Missing:

Continuous Learning / Runtime Validation

You discussed it extensively but roadmap barely reflects it.

Need explicit deliverable.

Recommended Addition

D10 — Runtime Validation & Continuous Learning

Tasks

- predicted vs actual comparison
- model health monitoring
- drift analysis
- retraining trigger logic
- runtime model validation
- continuous model improvement

This is VERY important because it differentiates:

- static SI
vs
 - intelligent HDT framework
-

Final Overall Assessment

Area

Status

Architecture flow	Very good
Technical depth	Strong
Deliverable sequencing	Good
Numbering consistency	Needs cleanup
Runtime validation	Missing
D4 actual data phase	Incomplete
VC ↔ DT linkage	Missing
Repetition handling	Needs refinement
Industrial realism	Strong

Overall:

~80–85% mature roadmap already

Very solid for an R&T planning sheet.

can you give the list of tasks with little more information not just as bullet points and give the updated list

Below is the updated and refined task structure for WP5.3 with:

- repetitions removed
- architecture aligned

- clearer task descriptions
- proper separation between WP5.1 and WP5.3 responsibilities
- more engineering-oriented wording

This should be much more suitable for:

- roadmap reviews
- management discussions
- architecture reviews
- Jira/Epic decomposition
- TRL planning

Based on your uploaded planning sheet.

D0 — Requirement Gathering

D0.1 Gather HDT, VC and Referencing Requirements

Tasks

- Conduct workshops with OMTC and stakeholders to understand expected Hybrid Digital Twin (HDT), Virtual Commissioning (VC), and referencing capabilities.
- Identify expected process monitoring, prediction, optimization, and correction use cases.
- Collect constraints related to machine architecture, process flow, sensors, and deployment environment.
- Identify expected KPIs, defect indicators, and operational pain points.

Output

- Initial requirement specification
 - Use-case definition
 - Stakeholder alignment report
-

D0.2 Validation of Requirements with OMTC

Tasks

- Review gathered requirements with OMTC and domain experts.

- Validate technical feasibility and scope boundaries.
- Finalize project priorities and deliverable expectations.
- Align terminology, KPIs, and operational definitions across teams.

Output

- Approved requirement baseline
-

D1 — Hybrid Digital Twin Architecture & ICD

D1.1 Overall System Architecture Definition

Tasks

- Define the end-to-end Hybrid Digital Twin workflow:

data → behavioral model → simulation → optimization → correction → feedback
- Define interactions between WP5.1, WP5.2, MES, VC, referencing, and optimization modules.
- Define high-level system boundaries and responsibilities.
- Define runtime and offline execution flows.

Output

- High-level HDT architecture
-

D1.2 Functional Block Design

Tasks

- Define major functional modules:
 - System Identification
 - Behavioral Digital Twin
 - Simulation Framework

- Optimization Framework
 - Referencing
 - NC Correction
 - MES Integration
 - Virtual Commissioning
- Define responsibilities and interfaces of each block.
- Define expected inputs/outputs for each module.

Output

- Functional architecture document
-

D1.3 Data Flow & Interface Definition (ICD)

Tasks

- Define how data moves between:
 - WP5.1
 - SI models
 - Digital Twin
 - MES
 - VC
- Define interface contracts and data exchange mechanisms.
- Define metadata structure and communication patterns.
- Define traceability requirements for exchanged data.

Output

- Interface Control Document (ICD)
-

D1.4 Referencing Integration Architecture

Tasks

- Define how referencing/localization outputs update NC correction and DT environments.
- Define coordinate transformation workflow.
- Define integration with external metrology systems.

Output

- Referencing integration workflow
-

D1.5 NC Correction Strategy Definition

Tasks

- Define static NC correction methodology:

OLP → New Reference Frame → Corrected NC
- Define transformation logic for TCP positions and orientations.
- Define corrected NC output interfaces.

Output

- NC correction strategy document
-

D1.6 MES Integration Architecture

Tasks

- Define MES interaction points.
- Define process traceability workflow.
- Define optimization and DT metadata storage strategy.

Output

- MES integration architecture
-

D1.7 Virtual Commissioning Integration

Tasks

- Define how VC validates:
 - behavioral DT outputs
 - optimized parameters
 - NC corrections

- Define simulation-to-deployment validation workflow.

Output

- VC integration workflow
-

D2 — Modeling Data Preparation (Sample mTorres Data)

D2.1 Identify Required Process Signals & KPIs

Tasks

- Identify process variables relevant for modeling.
- Define process KPIs influencing quality and defects.
- Identify machine, sensor, and contextual signals.
- Identify relationships between process behavior and defects.

Output

- Initial process signal list
 - KPI definition document
-

D2.2 Collect Historical/Sample Machine Data

Tasks

- Gather available mTorres sample datasets.
- Organize process, machine, and sensor datasets.
- Review process variability and available defect information.
- Build initial data repository for SI activities.

Output

- Initial sample dataset repository

D2.3 Define SI Modeling Variables (Planning Phase)

Tasks

- Define model inputs and prediction outputs.
- Define target process behaviors to be represented by the SI model.
- Define process operating states:
 - Laydown
 - Travel
 - Cut/Add
 - Retract
- Define behavioral labels and state transitions.
- Validate KPIs and prediction targets with stakeholders.

Output

- SI variable mapping
 - Behavioral state definition
-

D2.4 Modeling Dataset Structuring

Tasks

- Select signals relevant for behavioral modeling.
- Prepare time-sequence/window-based datasets.
- Organize datasets into:
 - train
 - validation
 - test
- Group datasets by manufacturing scenarios and operating conditions.
- Format datasets for MATLAB/Python SI frameworks.

Output

- Structured SI-ready datasets
-

D2.5 Feature & Label Preparation for SI

Tasks

- Generate derived features useful for modeling.
- Align defect and event labels with process timelines.
- Prepare behavioral state classifications.
- Balance datasets to improve model robustness.
- Align delayed variables and time-dependent process effects.

Output

- Feature-engineered labeled datasets
-

D2.6 Dataset Validation for SI Readiness

Tasks

- Validate dataset completeness and coverage.
- Validate compatibility with SI workflows.
- Validate coverage across different process conditions.
- Release curated dataset for initial SI activities.

Output

- Validated SI-ready dataset
-

D3 — Data Ingestion from WP5.1 — Design & Development

D3.1 Interface Contract Definition

Tasks

- Define schema for data exchange between WP5.1 and WP5.3.
- Define signal naming conventions, metadata, units, and timestamps.

- Define ingestion frequency and synchronization expectations.
- Define storage and traceability structure.

Output

- WP5.1 ↔ WP5.3 data interface specification
-

D3.2 Data Ingestion Pipeline Development

Tasks

- Develop ingestion framework/API for WP5.1 datasets.
- Develop dataset parsers and validation logic.
- Implement dataset registration and versioning.
- Build storage structure for SI and DT workflows.
- Develop sample-data simulator adapter for pipeline testing.

Output

- Reusable ingestion pipeline
-

D3.3 WP5.1 Integration Readiness Validation

Tasks

- Validate ingestion workflow using mock/sample datasets.
- Validate schema compatibility and extensibility.
- Perform dry-run integration testing.
- Validate ingestion stability and traceability.

Output

- Validated ingestion architecture
-

D4 — Actual Laying Data Integration & Modeling Preparation

D4.1 Actual Laying Data Reception

Tasks

- Receive curated laying datasets from WP5.1.
- Import datasets through developed ingestion pipeline.
- Validate dataset completeness and metadata availability.
- Register and organize incoming datasets.

Output

- Integrated actual laying datasets
-

D4.2 SI Variable Refinement using Actual Laying Data

Tasks

- Validate and refine SI inputs/outputs using real production data.
- Refine process states and behavioral labels.
- Refine KPI definitions and prediction targets.
- Validate process-state segmentation against actual process behavior.

Output

- Refined behavioral modeling variables
-

D4.3 Modeling Dataset Preparation

Tasks

- Prepare SI-ready datasets using actual laying data.
- Generate scenario-based and sequence-based datasets.
- Prepare datasets for DT simulation and optimization workflows.

- Organize datasets by process conditions and defect classes.

Output

- SI/DT-ready structured datasets
-

D4.4 Defect-to-Process Mapping

Tasks

- Link defects to corresponding process windows and trajectories.
- Map process conditions to defect occurrence.
- Create initial root-cause relationship mappings.

Output

- Defect-process correlation dataset
-

D4.5 Dataset Validation for SI & DT Usage

Tasks

- Validate dataset compatibility with SI and DT workflows.
- Validate prediction target availability.
- Validate scenario coverage and operational diversity.

Output

- Validated production-ready modeling dataset
-

(continued in your later deliverables: SI, DT, Optimization, VC, Runtime Validation, MES, Referencing)

can you give the list of tasks again with approximate days or months required to complete it

D0 — Requirement Gathering

Total Duration: ~2.5 Months

Task	Approx Duration
D0.1 Gather HDT, VC and referencing requirements	6–7 weeks
D0.2 Requirement validation with OMTC	2 weeks

D1 — Hybrid Digital Twin Architecture & ICD

Total Duration: ~5 Months

Task	Approx Duration
D1.1 Overall system architecture definition	3 weeks
D1.2 Functional block design	6–8 weeks
D1.3 Data flow & interface definition (ICD)	8–10 weeks
D1.4 Referencing integration architecture	2–3 weeks
D1.5 NC correction strategy definition	2–3 weeks

D1.6 MES integration architecture	2 weeks
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D1.7 Virtual commissioning integration definition	2–3 weeks
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D2 — Modeling Data Preparation (Sample mTorres Data)

Total Duration: ~2 Months

Task	Approx Duration
D2.1 Identify required process signals & KPIs	2 weeks
D2.2 Collect historical/sample machine data	2 weeks
D2.3 Define SI modeling variables & behavioral states	2–3 weeks
D2.4 Modeling dataset structuring	2 weeks
D2.5 Feature & label preparation for SI	3–4 weeks
D2.6 Dataset validation for SI readiness	2 weeks

D3 — Data Ingestion from WP5.1 — Design & Development

Total Duration: ~2.5 Months

Task	Approx Duration
D3.1 Interface contract definition	3 weeks
D3.2 Data ingestion pipeline development	6–7 weeks
D3.3 WP5.1 integration readiness validation	2–3 weeks

D4 — Actual Laying Data Integration & Modeling Preparation

Total Duration: ~3.5 Months

Task	Approx Duration
D4.1 Actual laying data reception & integration	2 weeks
D4.2 SI variable refinement using actual laying data	3 weeks

D4.3 Modeling dataset preparation	4 weeks
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D4.4 Defect-to-process mapping	3 weeks
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D4.5 Dataset validation for SI & DT usage	2 weeks
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D5 — Ver1 System Identification (Sample Data / POC Phase)

Total Duration: ~4 Months

Task	Approx Duration
D5.1 System I/O mapping & KPI categorization	2 weeks
D5.2 Synthetic proxy plant / MATLAB-Python environment setup	3 weeks
D5.3 Implement SI algorithms	5–6 weeks
D5.4 Initial model training & calibration	3 weeks
D5.5 Offline validation loop 1 (train/test validation)	3 weeks
D5.6 Residual analysis & model refinement	2 weeks

D5.7 Initial behavioral model release

1 week

D6 — Ver2 System Identification Improvement with WP5.1 Data

Total Duration: ~2.5 Months

Task	Approx Duration
D6.1 Input-output mapping refinement	2 weeks
D6.2 Model structure refinement	2 weeks
D6.3 Model retraining using actual laying data	3 weeks
D6.4 Model validation & residual analysis	2 weeks
D6.5 Runtime drift analysis & monitoring	2 weeks

D7 — Ver1 Analysis Framework & Digital Twin (Sample Data)

Total Duration: ~5.5 Months

Task	Approx Duration
D7.1 Behavioral DT integration	4 weeks
D7.2 Scenario simulation framework	4 weeks
D7.3 Root cause analysis framework	3 weeks
D7.4 KPI correlation framework	3 weeks
D7.5 Sensitivity analysis framework	3 weeks
D7.6 Predicted output visualization framework	2 weeks
D7.7 DT validation using sample scenarios	2 weeks

D8 — Ver2 Analysis Framework & DT using Actual WP5.1 Data

Total Duration: ~1.5–2 Months

Task	Approx Duration
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D8.1 DT refinement using actual laying data	2 weeks
D8.2 Validation of KPI correlation framework	2 weeks
D8.3 Validation of root cause framework	2 weeks
D8.4 Validation of sensitivity analysis	1–2 weeks
D8.5 Validation of predicted outputs	1–2 weeks

D9 — Visualization Integration with WP5.1

Total Duration: ~3.5 Months

Task	Approx Duration
D9.1 Define visualization interface	2 weeks
D9.2 Predicted output integration with WP5.1	4 weeks
D9.3 Model health & KPI visualization	3 weeks
D9.4 Dashboard integration support	3 weeks

D9.5 Validation with WP5.1

2 weeks

D10 — Optimization Framework (Sample Data)

Total Duration: ~4.5 Months

Task	Approx Duration
D10.1 Optimization objective definition	2 weeks
D10.2 Constraint framework definition	3 weeks
D10.3 Optimization engine development	5 weeks
D10.4 Parameter recommendation logic	3 weeks
D10.5 Optimization studies using sample data	3 weeks
D10.6 Validation of optimization workflow	2 weeks

D11 — Optimization Framework with Actual Laying Data

Total Duration: ~2 Months

Task	Approx Duration
D11.1 Safe operating window definition	2 weeks
D11.2 Optimization refinement using real data	3 weeks
D11.3 Validation of optimization recommendations	2 weeks
D11.4 Final optimization framework release	1 week

D12 — Runtime Prediction & Continuous Learning

Total Duration: ~3 Months

Task	Approx Duration
D12.1 Real-time defect prediction framework	4 weeks

D12.2 Deviation detection framework	3 weeks
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D12.3 Early warning indicator framework	3 weeks
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D12.4 Runtime validation loop (Prediction vs Actual)	3 weeks
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D13 — Virtual Commissioning

Total Duration: ~15–16 Months

Task	Approx Duration
D13.1 VC architecture definition	4 months
D13.2 Simulation model creation	3–4 months
D13.3 Integration of simulation models	3 months
D13.4 VC testing & validation	2 months
D13.5 Shop floor integration support	5 months
D13.6 HDT + VC integration testing	3 months

D13.7 End-to-end integration with DAQ 3 months

D14 — Zero-Time Roto-Translation / Referencing

Total Duration: ~10 Months

Task	Approx Duration
D14.1 CAD feasibility & line-of-sight study	2–3 weeks
D14.2 Referencing system architecture definition	2 months
D14.3 Electrical design	3–4 weeks
D14.4 Procurement & launch readiness	1–2 months
D14.5 Mechanical structure & target support design	1–2 months
D14.6 Referencing software development	3 months
D14.7 Integration & POC	1–1.5 months
D14.8 Testing & validation	1 month

D14.9 Industrialization documentation 2 weeks

D14.10 Referencing output interface to NC correction 2 weeks

D15 — MES Integration & Feedback Loop

Total Duration: ~3–4 Months

Task	Approx Duration
D15.1 MES integration architecture	3 weeks
D15.2 MES integration implementation	6 weeks
D15.3 Traceability & metadata integration	2 weeks
D15.4 Workflow testing & operator validation	3 weeks
D15.5 Feedback loop validation	2 weeks

D5 — Ver1 System Identification (Sample Data / POC Phase)

Objective

Develop the initial behavioral model using sample mTorres datasets before real laying data becomes available from WP5.1.

This phase focuses on:

- proving feasibility
- developing SI workflows
- validating modeling approaches
- building the first behavioral model (Ver1)

Estimated Duration

~4 Months

D5.1 System I/O Mapping & KPI Categorization

Description

Define how machine/process variables interact and identify which variables act as model inputs, outputs, disturbances, or KPIs.

Activities

- Define process input variables:
 - temperature
 - pressure
 - laying speed
 - tension
 - TCP position
- Define target outputs:
 - defects
 - process quality
 - deviation
 - instability indicators

- Categorize KPIs into:
 - quality KPIs
 - process KPIs
 - machine KPIs
- Define disturbance variables and external influences.
- Define behavioral boundaries between different process states.

Outputs

- System input/output mapping
- KPI categorization document

Estimated Duration

~2 Weeks

D5.2 Synthetic Proxy Plant / Modeling Environment Setup

Description

Build the initial modeling environment and synthetic proxy plant required to test SI workflows before production-grade data becomes available.

Activities

- Setup MATLAB/Python SI environment.
- Define reusable modeling workflows.
- Develop synthetic/mock process behavior generators.
- Create baseline process simulation environment.
- Develop reusable training/evaluation scripts.
- Define experiment configuration structure.

Outputs

- SI development environment
- Synthetic proxy process model

Estimated Duration

~3 Weeks

D5.3 Implement System Identification Algorithms

Description

Implement and evaluate multiple System Identification approaches for modeling process behavior.

Activities

- Implement baseline SI methods:
 - ARX
 - State-space
 - regression-based models
- Evaluate ML-based modeling approaches:
 - neural networks
 - LSTM/time-series models
- Evaluate hybrid/physics-informed approaches.
- Define model training workflows.
- Define parameter estimation workflows.
- Compare model complexity vs prediction accuracy.

Outputs

- Initial behavioral model candidates
- SI framework implementation

Estimated Duration

~5–6 Weeks

D5.4 Initial Model Training & Calibration

Description

Train behavioral models using sample datasets and calibrate model parameters to represent process behavior accurately.

Activities

- Train SI models using prepared datasets.
- Calibrate model parameters.

- Tune model hyperparameters.
- Improve prediction stability.
- Validate transient and steady-state process behavior representation.
- Compare predicted vs expected process behavior.

Outputs

- Trained Ver1 behavioral models

Estimated Duration

~3 Weeks

D5.5 Offline Validation Loop 1

Description

Validate the behavioral model using offline train/test workflows before DT integration.

Activities

- Perform train/validation/test evaluation.
- Compute prediction accuracy metrics.
- Perform residual/error analysis.
- Evaluate process behavior prediction quality.
- Validate model robustness across scenarios.
- Evaluate model generalization capability.

Outputs

- Offline validation report
- Model accuracy metrics

Estimated Duration

~3 Weeks

D5.6 Residual Analysis & Model Refinement

Description

Improve the model based on validation errors and residual analysis.

Activities

- Analyze prediction residuals.
- Identify unstable operating regions.
- Refine feature selection.
- Improve model structure.
- Re-tune parameters and retrain if required.
- Improve prediction robustness.

Outputs

- Refined behavioral model

Estimated Duration

~2 Weeks

D5.7 Initial Behavioral Model Release

Description

Finalize the initial behavioral model for use inside the Behavioral Digital Twin environment.

Activities

- Freeze validated Ver1 model.
- Export reusable model artifacts.
- Document model assumptions and limitations.
- Prepare model interfaces for DT integration.

Outputs

- Ver1 behavioral model release

Estimated Duration

~1 Week

D6 — Ver2 System Identification Improvement using WP5.1 Data

Objective

Improve and refine the initial behavioral model using actual laying data from WP5.1.

Estimated Duration

~2.5 Months

D6.1 Input-Output Mapping Refinement

Description

Refine the original SI assumptions using real laying process behavior.

Activities

- Refine process variables and KPIs.
- Validate prediction targets using actual process behavior.
- Refine process-state definitions.
- Identify additional process dependencies.
- Improve disturbance modeling.

Outputs

- Refined SI variable mapping

Estimated Duration

~2 Weeks

D6.2 Model Structure Refinement

Description

Improve the structure and complexity of the behavioral model using real process behavior.

Activities

- Compare model structures using real data.
- Improve transient behavior representation.
- Improve nonlinear behavior modeling.
- Improve defect prediction capability.
- Evaluate simplified vs complex model tradeoffs.

Outputs

- Improved behavioral model structure

Estimated Duration

~2 Weeks

D6.3 Model Retraining using Actual Laying Data

Description

Retrain the behavioral model using actual production datasets from WP5.1.

Activities

- Retrain models using actual laying datasets.
- Improve parameter estimation.
- Improve behavioral prediction quality.
- Improve process-state prediction accuracy.
- Validate model stability using real process conditions.

Outputs

- Ver2 trained behavioral model

Estimated Duration

~3 Weeks

D6.4 Model Validation & Residual Analysis

Description

Validate the refined model using production-level process scenarios.

Activities

- Evaluate prediction accuracy.
- Compare predicted vs actual process behavior.
- Evaluate robustness across different laying conditions.
- Evaluate model residuals and drift.
- Validate defect prediction capability.

Outputs

- Ver2 validation report

Estimated Duration

~2 Weeks

D6.5 Runtime Drift Analysis & Monitoring

Description

Prepare the model for runtime validation and continuous learning workflows.

Activities

- Define prediction-vs-actual comparison workflows.
- Define drift detection strategy.
- Define runtime monitoring KPIs.
- Define retraining trigger logic.
- Define model health indicators.

Outputs

- Runtime validation framework

Estimated Duration

~2 Weeks

D7 — Ver1 Analysis Framework & Behavioral Digital Twin (Sample Data)

Objective

Develop the initial Behavioral Digital Twin and process intelligence framework using the validated Ver1 behavioral model.

Estimated Duration

~5–5.5 Months

D7.1 Behavioral Digital Twin Integration

Description

Integrate the validated behavioral model into a Digital Twin execution environment capable of simulating process behavior.

Activities

- Integrate Ver1 behavioral model into DT workflow.
- Define DT execution pipeline.
- Define simulation interfaces.
- Define process scenario execution workflow.
- Define prediction output structure.

Outputs

- Ver1 Behavioral Digital Twin

Estimated Duration

~4 Weeks

D7.2 Scenario Simulation Framework

Description

Develop capability to simulate virtual process scenarios and evaluate process behavior under varying conditions.

Activities

- Develop what-if simulation workflows.
- Simulate parameter variation studies.
- Simulate different process operating conditions.
- Simulate defect-prone operating regions.
- Generate predicted process outputs.

Outputs

- Scenario simulation framework

Estimated Duration

~4 Weeks

D7.3 Root Cause Analysis Framework

Description

Develop framework to identify relationships between defects and process parameters.

Activities

- Analyze defect-process relationships.
- Identify dominant defect contributors.
- Map process states to defect occurrence.
- Generate root-cause indicators.

Outputs

- Root-cause analysis framework

Estimated Duration

~3 Weeks

D7.4 KPI Correlation Framework

Description

Develop framework to analyze interactions between process KPIs and machine/process variables.

Activities

- Compute KPI relationships.
- Analyze process parameter interactions.
- Identify coupled process behavior.
- Generate KPI dependency maps.

Outputs

- KPI correlation framework

Estimated Duration

~3 Weeks

D7.5 Sensitivity Analysis Framework

Description

Develop framework to evaluate how process variations affect process quality and defects.

Activities

- Evaluate sensitivity to parameter changes.
- Identify dominant process variables.
- Evaluate process stability regions.

- Generate parameter influence maps.

Outputs

- Sensitivity analysis workflow

Estimated Duration

~3 Weeks

D7.6 Predicted Output Visualization Framework

Description

Develop interfaces to visualize DT predictions and analysis outputs.

Activities

- Generate predicted KPI outputs.
- Visualize predicted process behavior.
- Visualize process deviations and trends.
- Prepare outputs for WP5.1 visualization integration.

Outputs

- Predicted output visualization framework

Estimated Duration

~2 Weeks

D7 — Ver2 Analysis Framework & Behavioral Digital Twin (Using Actual WP5.1 Data)

Objective

Refine and validate the Behavioral Digital Twin and process intelligence framework using actual laying process datasets received from WP5.1. This phase focuses on improving realism, validating prediction quality, and refining engineering insights using real production behavior.

Estimated Duration

~1.5–2 Months

D7.1 Digital Twin Simulation (Process Behavior Prediction)

Description

Use the validated Ver2 behavioral model to simulate realistic process behavior under varying operating conditions and evaluate predicted quality and defect trends.

Activities

- Integrate Ver2 behavioral model into the DT execution environment.
- Simulate process behavior under varying:
 - laying speed
 - tension
 - temperature
 - pressure
 - geometric conditions
- Simulate defect-prone operating regions.
- Evaluate predicted process quality and instability behavior.
- Compare simulated behavior against actual laying data.

Outputs

- Refined Behavioral Digital Twin
- Realistic process behavior prediction capability

Estimated Duration

~3 Weeks

D7.2 Root Cause Analysis Framework Validation

Description

Refine and validate the framework that identifies relationships between process conditions and defect occurrence using actual laying data.

Activities

- Validate process-defect relationships using production datasets.
- Refine root-cause indicators.
- Identify dominant process variables influencing defects.
- Validate behavioral dependencies between process states and defects.
- Improve interpretability of process insights.

Outputs

- Refined root-cause analysis framework

Estimated Duration

~2 Weeks

D7.3 KPI Correlation Framework Validation

Description

Validate and improve KPI interaction analysis using actual production process behavior.

Activities

- Analyze relationships between machine/process KPIs.

- Identify coupled process behaviors.
- Refine KPI dependency models.
- Validate KPI interactions across multiple operating conditions.
- Improve process insight generation logic.

Outputs

- Validated KPI correlation framework

Estimated Duration

~2 Weeks

D7.4 Sensitivity Analysis Framework Validation

Description

Validate the sensitivity analysis workflow using actual laying process scenarios.

Activities

- Evaluate influence of parameter variations on:
 - quality
 - defects
 - process stability
- Identify dominant process sensitivities.
- Refine process stability boundaries.
- Validate sensitivity trends against production data.

Outputs

- Refined sensitivity analysis framework

Estimated Duration

~2 Weeks

D8 — Visualization of Predicted Outputs — Integration with WP5.1

Objective

Provide interfaces for sending DT predictions, KPI insights, model-health indicators, and optimization outputs to WP5.1 for monitoring and visualization.

Estimated Duration

~3–4 Months

D8.1 Visualization Interface Definition

Description

Define how predicted outputs and analysis results are transferred to WP5.1 visualization components.

Activities

- Define prediction output schema.
- Define visualization data structures.
- Define metadata and traceability information.
- Define update frequency and interface requirements.
- Define KPI and model-health indicators.

Outputs

- Visualization interface specification

Estimated Duration

~2 Weeks

D8.2 Predicted Output Integration with WP5.1

Description

Integrate DT prediction outputs into WP5.1 monitoring and visualization workflows.

Activities

- Export predicted process outputs.
- Export predicted machine behavior.
- Export predicted KPI trends.
- Export predicted defect indicators.
- Enable real-time or batch prediction exchange workflows.

Outputs

- WP5.1 visualization-ready prediction pipeline

Estimated Duration

~4 Weeks

D8.3 Model Health & Insight Visualization

Description

Provide indicators that allow WP5.1 to visualize model confidence and process intelligence outputs.

Activities

- Export model confidence indicators.
- Export prediction quality metrics.
- Export drift indicators.
- Export KPI insight summaries.
- Export optimization recommendations.

Outputs

- Model health and insight visualization interface

Estimated Duration

~3 Weeks

D8.4 Dashboard Integration & Validation

Description

Validate that prediction outputs and insights are correctly visualized inside WP5.1 monitoring components.

Activities

- Validate interface integration.
- Validate visualization refresh/update behavior.
- Validate predicted vs actual visualization workflows.
- Validate operational usability with stakeholders.

Outputs

- Validated visualization integration

Estimated Duration

~2 Weeks

D9 — Ver1 Optimization Framework (Using Sample Data)

Objective

Develop the initial optimization framework using the Ver1 Behavioral Digital Twin and sample datasets.

Estimated Duration

~4–4.5 Months

D9.1 Optimization Objective Definition

Description

Define optimization goals and engineering objectives for the AFP process.

Activities

- Define quality optimization objectives.
- Define defect minimization objectives.
- Define cycle-time optimization goals.
- Define multi-objective optimization priorities.
- Define optimization KPIs.

Outputs

- Optimization objective specification

Estimated Duration

~2 Weeks

D9.2 Constraint Framework Definition

Description

Define operational and physical constraints governing optimization workflows.

Activities

- Define machine limits.
- Define process safety constraints.
- Define feasible operating windows.
- Define geometric and process restrictions.
- Define optimization feasibility conditions.

Outputs

- Optimization constraint framework

Estimated Duration

~3 Weeks

D9.3 Optimization Engine Development

Description

Develop the optimization workflow capable of exploring process parameter combinations using DT predictions.

Activities

- Implement optimization algorithms.
- Integrate optimization with DT simulation environment.
- Develop parameter search workflows.
- Develop optimization evaluation metrics.
- Implement multi-objective optimization logic.

Outputs

- Ver1 optimization engine

Estimated Duration

~5 Weeks

D9.4 Parameter Recommendation Logic

Description

Develop logic for generating process parameter recommendations from optimization studies.

Activities

- Generate recommended process settings.
- Define safe parameter recommendations.
- Define process improvement suggestions.
- Define optimization reporting structure.

Outputs

- Parameter recommendation framework

Estimated Duration

~3 Weeks

D9.5 Optimization Validation using Sample Data

Description

Validate optimization behavior using simulated/sample process scenarios.

Activities

- Execute optimization studies.
- Validate optimization stability.
- Validate recommended parameter consistency.
- Evaluate optimization effectiveness.

Outputs

- Validated Ver1 optimization workflow

Estimated Duration

~2 Weeks

D10 — Ver2 Optimization Framework (Using Actual Laying Data)

Objective

Improve and validate the optimization framework using actual production datasets and refined DT models.

Estimated Duration

~2 Months

D10.1 Safe Operating Window Definition

Description

Define stable operating regions that ensure reliable and defect-minimized process execution.

Activities

- Analyze stable process operating conditions.
- Define acceptable parameter ranges.
- Identify instability regions.
- Define operational boundaries.

Outputs

- Safe operating window definition

Estimated Duration

~2 Weeks

D10.2 Optimization Engine Refinement using Actual Data

Description

Improve optimization behavior using refined DT predictions and actual process behavior.

Activities

- Refine optimization objectives using real process data.
- Improve optimization convergence.
- Improve recommendation quality.
- Validate optimization robustness.

Outputs

- Refined optimization framework

Estimated Duration

~3 Weeks

D10.3 Optimization Validation using Actual Laying Data

Description

Validate optimization recommendations using actual production scenarios.

Activities

- Execute optimization studies using actual datasets.
- Validate feasibility of recommendations.
- Validate process improvement potential.
- Validate process-stability recommendations.

Outputs

- Production-ready optimization framework

Estimated Duration

~2 Weeks

D11 — Real-Time Prediction & Runtime Validation Framework

Objective

Develop runtime prediction, deviation detection, and continuous-learning workflows using live process data.

Estimated Duration

~3 Months

D11.1 Real-Time Defect Prediction

Description

Develop runtime prediction workflows capable of estimating defect occurrence during process execution.

Activities

- Integrate runtime process streams with trained models.
- Generate real-time defect predictions.
- Generate runtime quality indicators.
- Validate runtime prediction latency and stability.

Outputs

- Runtime defect prediction framework

Estimated Duration

~4 Weeks

D11.2 Deviation Detection Framework

Description

Detect abnormal process behavior by comparing predicted and actual process outputs.

Activities

- Compare predicted vs actual process behavior.
- Detect process anomalies and deviations.
- Compute runtime prediction errors.
- Generate deviation indicators.

Outputs

- Runtime deviation detection framework

Estimated Duration

~3 Weeks

D11.3 Early Warning Indicator Framework

Description

Develop predictive indicators capable of identifying early instability trends before failures or defects occur.

Activities

- Analyze historical defect progression behavior.
- Define early warning indicators.
- Detect abnormal process evolution trends.
- Generate process instability indicators.

Outputs

- Predictive warning indicator framework

Estimated Duration

~3 Weeks

D11.4 Runtime Validation & Continuous Learning

Description

Continuously improve the behavioral model using runtime prediction feedback.

Activities

- Monitor prediction accuracy over time.
- Detect model drift.
- Define retraining triggers.
- Define continuous-learning workflows.
- Improve model robustness using runtime data.

Outputs

- Runtime validation and continuous-learning framework

Estimated Duration

~3 Weeks

what about virtual commissioning?

D12 — Virtual Commissioning Architecture

Objective

Define the overall Virtual Commissioning (VC) strategy, architecture, scope, and validation workflow for the AFP process and Hybrid Digital Twin ecosystem.

Estimated Duration

~3.5–4 Months

D12.1 VC Use-Case & Scope Definition

Description

Identify the Virtual Commissioning use cases and define which parts of the AFP system will be virtually modeled and validated before deployment.

Activities

- Identify VC objectives:
 - process validation
 - PLC validation
 - robot validation
 - safety validation
 - defect scenario testing
- Define VC scope:

- robot
- end-effector
- PLC
- safety systems
- process simulation
- Identify required simulation fidelity.
- Align VC expectations with stakeholders.

Outputs

- VC scope definition
- VC use-case specification

Estimated Duration

~3 Weeks

D12.2 VC Software & Hardware Architecture Definition

Description

Define the software stack, hardware interfaces, and communication architecture required for VC implementation.

Activities

- Define HiL/SiL architecture.
- Define communication interfaces.
- Define PLC/safety integration strategy.
- Define robot-controller interaction workflows.
- Define synchronization mechanisms between VC and DT.

Outputs

- VC architecture document

Estimated Duration

~4 Weeks

D12.3 VC Environment Setup

Description

Prepare the base simulation environment required for VC execution.

Activities

- Setup simulation tools/platforms.
- Configure communication interfaces.
- Setup robot and PLC simulation environments.
- Configure process simulation interfaces.
- Configure data exchange interfaces.

Outputs

- VC development environment

Estimated Duration

~4 Weeks

D13 — Simulation Model Creation (Robot, EE, PLC & Safety)

Objective

Develop virtual simulation models representing robot behavior, end-effector behavior, PLC logic, safety interactions, and process behavior.

Estimated Duration

~3–4 Months

D13.1 Robot Simulation Model Development

Description

Develop robot simulation models representing motion behavior and execution logic.

Activities

- Create robot kinematic models.
- Configure motion execution workflows.
- Integrate robot controller interfaces.
- Validate robot motion behavior.

Outputs

- Robot simulation model

Estimated Duration

~4 Weeks

D13.2 End-Effector & Process Model Development

Description

Develop simulation models representing AFP head/process behavior.

Activities

- Model AFP head behavior.
- Model process-state transitions.
- Simulate process events:
 - laydown
 - cut/add
 - retract
- Define process parameter interactions.

Outputs

- AFP process simulation model

Estimated Duration

~4 Weeks

D13.3 PLC & Safety Simulation Development

Description

Develop virtual PLC and safety behavior required for VC validation.

Activities

- Simulate PLC sequences.
- Simulate safety logic and interlocks.
- Simulate emergency and abnormal conditions.
- Validate controller interaction behavior.

Outputs

- PLC & safety simulation models

Estimated Duration

~4 Weeks

D13.4 Interface & Behavior Validation

Description

Validate interactions between robot, PLC, safety, and process models.

Activities

- Validate communication interfaces.
- Validate synchronization behavior.
- Validate process-state transitions.
- Validate command-response timing.

Outputs

- Integrated simulation model validation

Estimated Duration

~2 Weeks

D14 — Integration of Simulation Models

Objective

Integrate robot, process, PLC, safety, and DT components into a unified VC workflow.

Estimated Duration

~3 Months

D14.1 Integrated Scenario Execution Framework

Description

Develop workflows for executing integrated virtual process scenarios.

Activities

- Define integrated test scenarios.
- Integrate robot + PLC + DT workflows.
- Integrate process-event simulation.
- Integrate defect simulation scenarios.

Outputs

- Integrated VC execution framework

Estimated Duration

~4 Weeks

D14.2 Scenario Testing & Validation

Description

Validate virtual process execution under multiple operational scenarios.

Activities

- Execute normal production scenarios.
- Execute defect scenarios.
- Execute abnormal operating conditions.
- Evaluate process behavior and responses.

Outputs

- Scenario validation reports

Estimated Duration

~4 Weeks

D14.3 HDT + VC Interaction Validation

Description

Validate interaction between Behavioral DT predictions and VC workflows.

Activities

- Validate DT predictions inside VC.
- Validate optimized parameter behavior in VC.
- Validate predicted defect scenarios.
- Validate NC correction behavior.

Outputs

- HDT + VC integration validation

Estimated Duration

~3 Weeks

D15 — Virtual Commissioning Testing & Validation

Objective

Validate readiness of the VC environment before deployment and shop-floor integration.

Estimated Duration

~2–2.5 Months

D15.1 Pre-Deployment Validation

Description

Validate process execution and system behavior before physical deployment.

Activities

- Validate integrated workflows.
- Validate process execution stability.
- Validate safety behavior.
- Validate process synchronization.

Outputs

- Pre-deployment validation report

Estimated Duration

~4 Weeks

D15.2 Defect & Failure Scenario Validation

Description

Validate VC capability to simulate abnormal process conditions and defect behavior.

Activities

- Simulate process deviations.
- Simulate defect conditions.
- Validate alarm/response behavior.
- Validate operational recovery scenarios.

Outputs

- Defect scenario validation report

Estimated Duration

~3 Weeks

D15.3 Performance & Readiness Assessment

Description

Assess whether the VC environment is sufficiently stable and representative for deployment support.

Activities

- Evaluate VC stability.
- Evaluate simulation fidelity.
- Evaluate synchronization accuracy.
- Document limitations and improvement areas.

Outputs

- VC readiness assessment

Estimated Duration

~2 Weeks

D16 — Shop-Floor Integration & Deployment Support

Objective

Support integration of VC and HDT workflows with actual shop-floor systems.

Estimated Duration

~5 Months

D16.1 Shop-Floor Integration Preparation

Description

Prepare deployment interfaces and workflows required for integration with real systems.

Activities

- Define deployment workflows.
- Define runtime integration interfaces.
- Prepare operational support workflows.
- Prepare troubleshooting strategy.

Outputs

- Deployment integration plan

Estimated Duration

~3 Weeks

D16.2 Shop-Floor Integration & Troubleshooting

Description

Support deployment and debugging activities during integration with actual production systems.

Activities

- Support runtime deployment.
- Troubleshoot integration issues.
- Validate communication interfaces.
- Validate process synchronization.

Outputs

- Integrated VC deployment support

Estimated Duration

~3 Months

D16.3 Operational Validation

Description

Validate VC-assisted workflows under real operational conditions.

Activities

- Validate operational process execution.
- Validate runtime process behavior.
- Validate operational usability.
- Collect operational improvement feedback.

Outputs

- Operational validation report

Estimated Duration

~1 Month

D17 — HDT + VC + DAQ End-to-End Integration

Objective

Perform full integration between Hybrid Digital Twin, Virtual Commissioning, and DAQ/process monitoring systems.

Estimated Duration

~3 Months

D17.1 HDT + VC Integration

Description

Integrate the Behavioral DT ecosystem with VC workflows.

Activities

- Integrate DT prediction interfaces.
- Integrate optimization outputs.
- Integrate NC correction outputs.
- Validate runtime synchronization.

Outputs

- Integrated HDT + VC framework

Estimated Duration

~4 Weeks

D17.2 DAQ Integration

Description

Integrate runtime process acquisition and monitoring systems with HDT + VC workflows.

Activities

- Integrate DAQ interfaces.
- Validate process monitoring synchronization.
- Validate runtime prediction updates.
- Validate process replay capability.

Outputs

- Integrated DAQ workflow

Estimated Duration

~4 Weeks

D17.3 End-to-End Validation

Description

Validate complete workflow:

DAQ → SI → DT → Optimization → VC → Monitoring

Activities

- Execute end-to-end operational scenarios.
- Validate runtime synchronization.
- Validate prediction/update workflows.
- Validate full-system traceability.

Outputs

- End-to-end validation report

Estimated Duration

~3 Weeks